

Heart Region Extraction and Segmentation from Chest CT Images using Hopfield Artificial Neural Networks

Rachid Sammouda, Rami Mohammad Jomaa, Hassan Mathkour
 Department of Computer Science, College of Computer and Information Science
 King Saud University
 Riyadh, Saudi Arabia
 sammoudar@yahoo.com, rami.m.j@hotmail.com, binmathkour@yahoo.com

ABSTRACT—A system for extracting and segmenting heart regions from three-dimensional (3D) CT chest images is proposed in this paper. At first, the regions of interest (ROIs) are extracted using pure basic image processing techniques applied on the 2D CT slices. Secondly, the ROIs in each slice are segmented using Hopfield Artificial Neural Networks (HANN). The segmentation results include tissues belonging to the heart and its surrounding organs. To distinguish between heart regions and the non-heart regions, a rule-based filtering approach is adopted. The system is evaluated using a database of 735 chest CT slices from 5 patients. It shows a good and accurate performance with some exceptions.

Keywords— 3D Chest CT images; Heart Region Extraction; Segmentation; and Hopfield Neural Networks

I. INTRODUCTION

With near to 17.3 million people dying from cardiovascular diseases (CVDs) in 2008, representing 30% of all death cases in the world, this percentage makes CVDs the leading cause of death in the world [1]. Thus, special efforts are required and dedicated to early diagnosis and treatment of cardiac diseases. 3D imaging is now a well-developed model for non-invasive diagnosis of cardiovascular disease. Computed tomography imaging modality was at a past time not a very popular option for cardiovascular tests [2]. However, in the last two decades, CT took the lead in comparison with other imaging modalities (such as magnetic resonance imaging MRI and ultra-sound), mainly because of its superior spatial resolution [3]. CT images can provide detailed anatomic information about the heart chambers, large vessels, and coronary arteries [4]. Therefore, CT is an important imaging modality for diagnosis of cardiovascular diseases. Multi-detector CT scanners produce a large amount of data (e.g. 300 slices per patient), which require long time for data interpretation. Consequently, an automatic extraction and segmentation of CT images is a pre-requisite for efficient handling and interpretation of this huge data. In this work, we introduce a method for automatic extraction and segmentation of the heart from chest CT images.

The next section includes an overview on our proposed method, then some results of our method are presented and discussed, and the conclusions are in the last section.

A. Related Work

Several algorithms have been achieved for the extraction and segmentation of the heart in the past [2]. Most of the earlier work was for the segmentation of the left or right ventricle or both because of their importance in the heart and whole body functionalities. The heart extraction and segmentation methods can be classified by many approaches. Some of methods work on fitting of geometric models to the medical images [5]. Another work on active shape models [6–8], active appearance models [9–12], and flexible deformable models allow local deformations subject to a smoothness constraint [13–15]. Lately, some approaches have been suggested for the segmentation of the whole heart and its chambers using techniques based on registration [12], [16], [17] or using geometric model [18], [19]. Some other methods work on the images using thresholds and clustering [20].

II. METHODS

The purpose of our work is to develop a method for automatic extraction of the heart based on the segmentation of the ROIs extracted from the chest CT images using artificial neural networks. Fig. 1 shows the proposed architecture of our method, where the first step is a pre-processing for the ROIs; this step involves primarily lungs borders extraction from the original chest CT images.

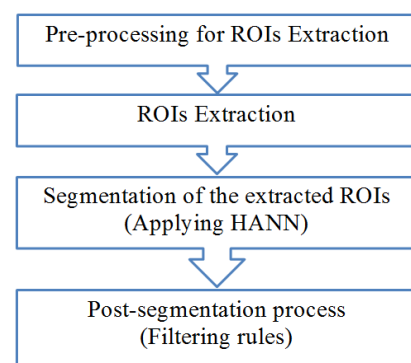


Figure 1. The proposed method (Automatic Heart Extraction System)

The next step of the proposed method is to extract the ROI from each slice. After we completely extract the ROIs, we start the segmentation step using a type of artificial neural networks named Hopfield ANN. By applying the segmentation process, different types of tissues located inside the ROIs (some of them belong to the heart and some to other organs) are classified. Our interest is in the clusters representing the heart tissues. To distinguish between the different resulted clusters, a post-segmentation process is applied. This process mainly involves formulating some filtering rules to eliminate the non-heart clusters from the segmentation result. Fig. 2 shows the data flow within our proposed system.

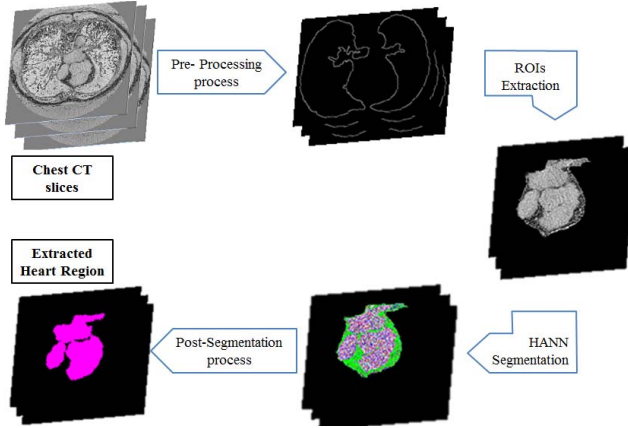


Figure 2. Data flow in the proposed system

B. Pre-processing for ROIs Extraction

In order to extract the ROIs from CT images, we do firstly a pre-processing step that includes automatic extraction of lungs borders. This step helps us in correctly determining the location of the heart. We based our work on previous processing techniques done on lung extraction used in CAD system [21]. Fig. 3 presents the pre-processing steps in order to extract the lungs borders. This process depends on applying some basic image processing techniques on the second bit-plane slice of each CT slice, these techniques include erosion, dilation, median filter, and outlining.

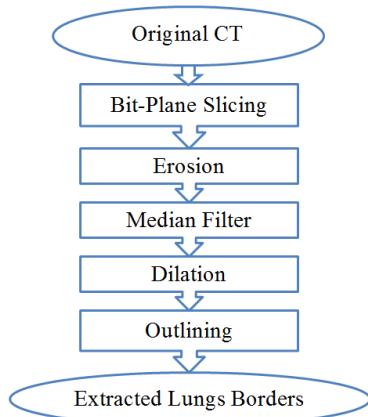


Figure 3. Pre-Processing step

C. Regions Of Interest (ROIs) Extraction

The second step in our proposed method is to extract the ROIs (which include the heart regions) from the CT slices. We depend on the extracted lungs borders achieved in the previous step. By applying some basic image processing techniques on the extracted lungs borders images, we can extract the ROIs located between the two lungs as shown in the Fig. 4. In this process, we search for the region located between the two lungs and consider it as the ROI. We can add another constraint on ROI extraction by removing the upper and lower parts of ROI, this constraint helps us concentrate on small ROI that contain the heart and hence reduces the amount of surrounding tissues included.

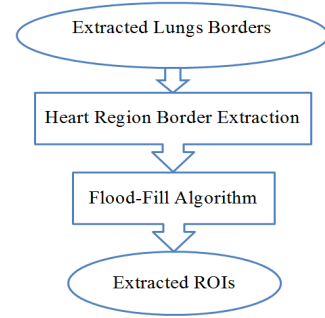


Figure 4. ROIs Extraction Process

D. Segmentation of the ROIs

After successfully extracting the ROIs from the CT slices, as described in the previous section, we can start the third step of the method, which is the segmentation of ROIs. This step aims at segmentation of the ROIs searching for tissues belonging to the heart. We use for this purpose artificial neural networks, which are well-known techniques used for many purposes and in several applications. ANNs are application independent and give good work with most of the applications. In our research we will use a known model of ANN namely Hopfield Artificial Neural Networks to solve our segmentation problem.

1) Hopfield Artificial Neural Networks (HANNs)

Hopfield neural network (HANN) is a known model of the artificial neural networks, which has been used in many research for different purposes. It has been applied in medical image processing, especially for classification of magnetic resonance (MR) images of the brain depending on minimization of the energy as described in [22], [23]. Both papers presented good results. In [22], they improved the algorithm that has been used in [23] to overcome some of the limits reported by its authors, such as minimizing the sum of squared errors and guaranteeing the network convergence in a pre-defined period. In our research, we use the improved version of the HANN. We formulate the segmentation of ROIs problem as an energy minimizing function constructed of a cost-term as a sum of squared errors. To ensure the network convergence, we achieve the minimization with a step function allowing the network to reach the stability stage in a pre-defined period. HANN classifier architecture is composite of a single layer representing a grid of $N \times M$ neurons, where N

represents the size of the given image and M represents the number of classes that is given as a priori information. Each column represents a class and each row represents a pixel. All neurons work as both input and output neurons at the same time. The neurons within each class hold values representing the probabilities that the corresponding pixel belongs to that class. The network is designed to classify the feature space without teacher using the distance measure between the pixel and the centroid of class. The segmentation problem is formulated as a partition of N pixels among M classes such that the assignment of the pixels minimizes the cost-term of the energy function:

$$E = \frac{1}{2} \sum_{k=1}^N \sum_{l=1}^M R_{kl} V_{kl}^2 \quad (1)$$

Where R_{kl} is the distance measure between the k^{th} pixel and the centroid of class l and V_{kl} is the output of the k^{th} neuron. By experiments, we found that hamming distance gives good and robust results; therefore we use it in this work. Hamming distance is determined as the number of the different bites between two words:

$$R_{kl} = \sum_{i=0}^{18} \mu(X_{kl}, \bar{X}_{kl}) \quad (2)$$

Where $\mu(a, b) = 1$ if $a \neq b$, 0 if $a = b$

In our case $n=2$ which means that the energy is defined as sum-squared error. The energy function is minimized using HANN by solving the motion equations that satisfy:

$$\frac{\partial U_{kl}}{\partial t} = -\mu(t) \frac{\partial E}{\partial V_{kl}} \quad (3)$$

Where U_{kl} and V_{kl} are the input and the output of the k^{th} neuron, $\mu(t)$ is a scalar positive function of time, which determines the length of the step to be taken in the direction of the vector $d = -\nabla E(V)$. The suitable choice of this step $\mu(t)$ requires some experience. In our case, by experiments, we utilized the function proposed in [22] for segmenting the MR data using HANN. We found it to be suitable for our work in segmenting of the CT images using the HANN:

$$\mu(t) = t(T_c - t) \quad (4)$$

Where t is the step of iteration and T_c is the pre-specified convergence time. HANN segmentation algorithm can be described as follows:

1. Initialize the input of neurons with random values.
2. Apply the input-output function V_{kl} to obtain the new output values for each neuron, which assigns the pixels to classes.

$$V_{kl}(t+1) = 1 \text{ if } U_{kl} = \text{Max}[U_{kl}(t), \forall l] \\ \text{else } V_{kl}(t+1) = 0 \quad (5)$$

3. For each class l , compute the centroid ($\bar{X}_l$) as follows:

$$\bar{X}_l = \frac{\sum_{k=1}^N X_{kl} V_{kl}}{n_l} \quad (6)$$

4. Compute the energy function (E) as defined in (1).
5. Update the inputs (U_{kl}) by solving the above deferential equation (3) using Euler's approximation. The learning operation occurs when neuron input weights are adjusted to minimize the output energy.

$$U_{kl}(t+1) = U_{kl}(t) + \frac{\partial U_{kl}}{\partial t} \quad (7)$$

$$1 \leq k \leq N, 1 \leq l \leq M$$

6. While $t < T_c$, then repeat from step 2. This process iteratively modifies the pixel assignments to reach a close optimal final segmentation map.

E. Post-Segmentation Process

After completing the segmentation process, we can distinguish between two types of objects inside the result; some of them belong to the heart and others to the surrounding organs and tissues. Our aim is to keep the objects representing the heart tissues and discard the others. To accomplish this end, we apply a post-segmentation process. This process involves two steps, the first one aims at smoothing the segmentation result, and the second includes formulating some filtering rules to eliminate the non-heart clusters form the segmentation result. Accordingly, we extract different features of the segmentation result to use them in the filtering step.

1) Segmentation Result Smoothing

The result of the segmentation includes many clusters as we said before, thus we concentrate on the highest density cluster as a cluster representing the heart. In addition, we found by experiments that there are two other clusters belonging to the heart and we merge them with the first cluster. Finally, we try to fill the holes inside the heart regions and remove the isolated and semi-isolated pixels. This process makes the result smoother and better for filtering by the filter rules in the next steps.

2) Feature Extraction

Different types of features were extracted in previous works depending on the methods used and the purposes of these features. Some extracted features like connectivity, uniformity and position were used in [24]. In [25] they extract features like circularity, size, and mean brightness of (ROIs). In other work [26] features like Area, thickness, circularity, intensity and variance are extracted. We apply some of these features to formulate the filtering rules in our system. We use many features to formulate the filtering rules like the area of the object, the maximum drawable circle (MDC) inside the object, and mean intensity value of the object.

The area of the object: We can get benefit from this feature by eliminating very small object (Area is less than a

thresholding value) and isolated pixels (considered as noise in the segmented image). By using this feature, we can eliminate a good number of non-heart regions, moreover using it guarantees reducing the computation time needed in the next filtering steps.

The maximum drawable circle (MDC): This operation begins to draw a circle starting from a point inside the object. This circle should fulfill the condition that all the pixels inside the circle belong to the object. We try all the pixels inside the object as starting drawing point. The operation starts to draw a one-pixel radius size circle starting from a point inside the object. If the operation succeeds, we increase the radius by one pixel and try to redraw the circle. We continue in this manner until the last radius exceeds the border of the object. Then we record the radius size of the previous drawing that meets the condition specifying that all the pixels inside the circle belong to the object as our MDC. We repeat the operation until it covers all the candidate objects. At the end, each candidate object saves its maximum drawable circle to be used in the filtering process to eliminate more non-heart regions.

3) Formulation of Filtering Rules

After extracting the above-mentioned features, we can formulate some filtering rules. Next, we refer to two rules:

Rule 1: For each candidate object, if the area of the object is less than a certain threshold T1 delete it from the candidate list. Applying this filter has the effect of decreasing the number of non-heart objects. This decision also reduces the computation time needed in the following filtering rules. We suppose, by experiments, that the threshold value is greater than or equal to 120 pixels, because the heart object is almost greater than this value.

Rule 2: For each candidate object, if the value of maximum drawable circle (MDC) is less than a certain threshold T2 delete it from the candidate list. T2 should be chosen carefully. This rule can delete most of non-heart objects. We suppose, by experiments, that the threshold value is greater than or equal to 320 pixels.

III. RESULTS

The proposed method has been tested on 735 slices from five patients with about 150 slices for each patient. We consider that the slices belonging to the heart are about 45 slices ranging from slice 60 to slice 105. The size of each slice is 512×512 pixels. Some parameters should be set before starting like the number of clusters considered inside the ROI. In our case we have tested many values but work in this paper is based on four clusters. Another parameter is T_s , the pre-specified convergence time for the segmentation process, we depend on previous study in this field [27], the authors in that study found that HANN need 120 iterations to give close-optimal solution. We show some results of our system, in the next two Fig. 5, 6. Moreover, we present the outputs of the different steps of our method.

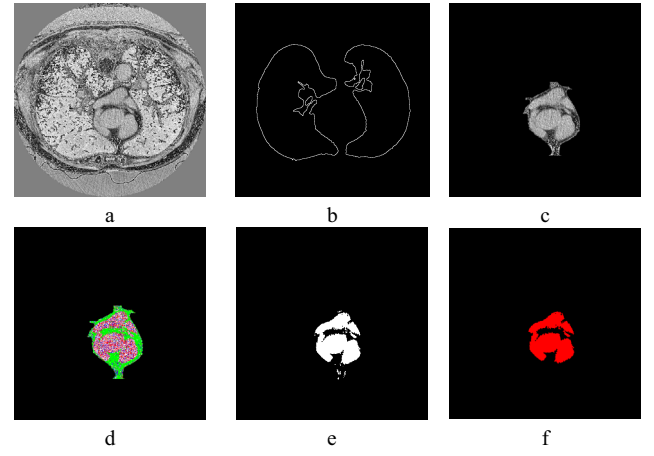


Figure 5. Heart Regions Extraction method: a. The original image (slice 70), b. Pre-processing, c. Extraction the ROIs, d. Applying HANN, e. Smoothing, f. Applying the filtering rules

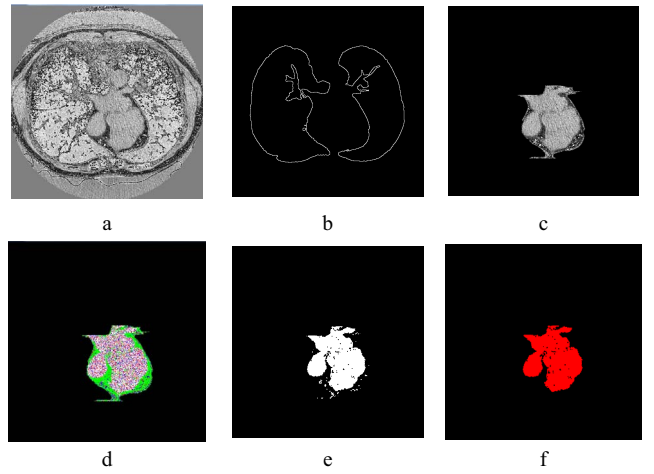
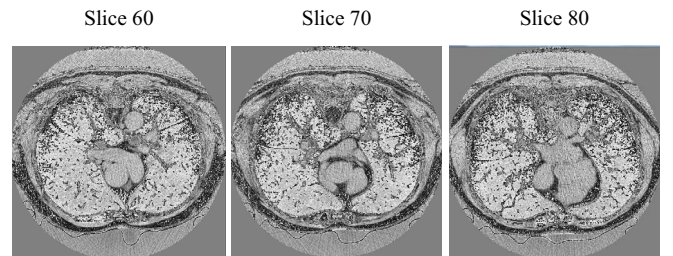


Figure 6. Heart Regions Extraction method: a. The original image (slice 80), b. Pre-processing, c. Extraction the ROIs, d. Applying HANN, e. Smoothing, f. Applying the filtering rules

In Fig. 7, we show the result of the method by applying it on different slices, while in Figure 8 we present sequence of slices with their results.



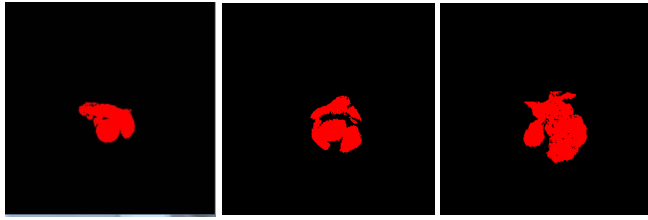


Figure 7. Heart Regions Extraction method (some slices): the first row represents the original slices, the second row represents the segmented heart regions

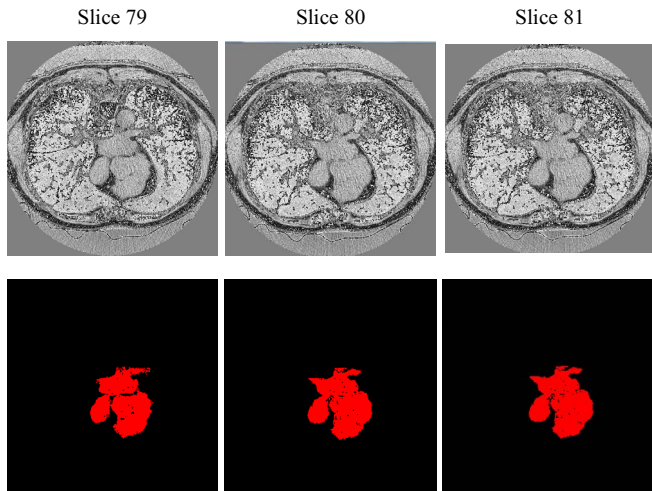


Figure 8. Heart Regions Extraction method (sequence of slices): the first row represents the original slices, the second row represents the segmented heart regions

One of the limitations of our method is that some very few slices have problems in the segmentation; these problems may be related to the higher contiguity between the heart and its neighbor tissues or because of the nature of data in the raw CT images. Other limitations, as shown in Fig. 9, are that some of holes still exist in the result heart or some small extra points appear in the heart, these may be because of the adhesion of the heart with the surrounding organs and sometimes because of the arteries that enter the lungs.

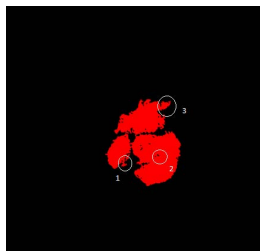


Figure 9. Some exceptions, 1 extra points, 2 holes inside the heart, 3 the arteries of the lungs

IV. CONCLUSIONS

We provide through this paper a robust and potential method for extraction of the heart from chest CT images. The

results show that our method gives good and accurate segmentation. Future work for our method could focus on increasing the performance and the accuracy of our method and applying it to other human organs.

V. ACKNOWLEDGES

The authors would like to thank the Research Center in the College of Computer and Information Sciences for their partial support for this work.

REFERENCES

- [1] "WHO | Cardiovascular diseases (CVDs)," *WHO*. [Online]. Available: <http://www.who.int/mediacentre/factsheets/fs317/en/index.html>. [Accessed: 27-Jan-2012].
- [2] A. F. Frangi, W. J. Niessen, and M. A. Viergever, "Three-dimensional modeling for functional analysis of cardiac images, a review," *Medical Imaging, IEEE Transactions on*, vol. 20, no. 1, pp. 2–5, 2001.
- [3] M. J. Budoff and J. S. Shinbane, Eds., *Cardiac CT Imaging: Diagnosis of Cardiovascular Disease*, 1st ed. Springer, 2006.
- [4] P. Schoenhagen, A. E. Stillman, S. S. Halliburton, and R. D. White, "CT of the heart: principles, advances, clinical uses," *Cleveland Clinic journal of medicine*, vol. 72, no. 2, pp. 127–138, 2005.
- [5] J. C. Cauvin, J. Y. Boire, M. Zanca, J. Maublan, and others, "3D modeling in myocardial 201TL SPECT," *Computerized medical imaging and graphics*, vol. 17, no. 4-5, pp. 345–350, 1993.
- [6] D. Fritz, D. Rinck, R. Dillmann, and M. Scheuering, "Segmentation of the left and right cardiac ventricle using a combined bi-temporal statistical model," in *Proc. of SPIE Medical Imaging*, 2006, pp. 605–614.
- [7] H. C. van Assen, M. G. Danilouchkine, M. S. Dirksen, J. H. C. Reiber, and B. P. F. Lelieveldt, "A 3-D active shape model driven by fuzzy inference: Application to cardiac CT and MR," *Information Technology in Biomedicine, IEEE Transactions on*, vol. 12, no. 5, pp. 595–605, 2008.
- [8] H. C. Van Assen et al., "SPASM: A 3D-ASM for segmentation of sparse and arbitrarily oriented cardiac MRI data," *Medical Image Analysis*, vol. 10, no. 2, pp. 286–303, 2006.
- [9] S. C. Mitchell, B. P. F. Lelieveldt, R. J. van der Geest, H. G. Bosch, J. Reiver, and M. Sonka, "Multistage hybrid active appearance model matching: segmentation of left and right ventricles in cardiac MR images," *Medical Imaging, IEEE Transactions on*, vol. 20, no. 5, pp. 415–423, 2001.
- [10] S. C. Mitchell, J. G. Bosch, B. P. F. Lelieveldt, R. J. van der Geest, J. H. C. Reiber, and M. Sonka, "3-D active appearance models: segmentation of cardiac MR and ultrasound images," *Medical Imaging, IEEE Transactions on*, vol. 21, no. 9, pp. 1167–1178, 2002.
- [11] M. B. Stegmann and D. Pedersen, "Bi-temporal 3D active appearance models with applications to unsupervised ejection fraction estimation," in *Proc. of SPIE, International Symposium on Medical Imaging*, 2005, vol. 5747, pp. 336–350.
- [12] X. Zhuang, K. S. Rhode, R. S. Razavi, D. J. Hawkes, and S. Ourselin, "A registration-based propagation framework for automatic whole heart segmentation of cardiac MRI," *Medical Imaging, IEEE Transactions on*, vol. 29, no. 9, pp. 1612–1625, 2010.
- [13] J. Montagnat and H. Delingette, "4D deformable models with temporal constraints: application to 4D cardiac image segmentation," *Medical Image Analysis*, vol. 9, no. 1, pp. 87–100, 2005.
- [14] M. R. Kaus, J. Berg, J. Weese, W. Niessen, and V. Pekar, "Automated segmentation of the left ventricle in cardiac MRI," *Medical Image Analysis*, vol. 8, no. 3, pp. 245–254, 2004.
- [15] T. McInerney and D. Terzopoulos, "A dynamic finite element surface model for segmentation and tracking in multidimensional medical images with application to cardiac 4D image analysis," *Computerized Medical Imaging and Graphics*, vol. 19, no. 1, pp. 69–83, 1995.

- [16] H. A. Kirisli et al., "Fully automatic cardiac segmentation from 3D CTA data: a multi-atlas based approach," in *Proc. of SPIE Vol.*, 2010, vol. 7623, pp. 762305–1.
- [17] J. Lötjönen, S. Kivistö, J. Koikkalainen, D. Smutek, and K. Lauerma, "Statistical shape model of atria, ventricles and epicardium from short-and long-axis MR images," *Medical image analysis*, vol. 8, no. 3, pp. 371–386, 2004.
- [18] Y. Zheng, A. Barbu, B. Georgescu, M. Scheuering, and D. Comaniciu, "Four-chamber heart modeling and automatic segmentation for 3-D cardiac CT volumes using marginal space learning and steerable features," *Medical Imaging, IEEE Transactions on*, vol. 27, no. 11, pp. 1668–1681, 2008.
- [19] O. Ecabert et al., "Automatic model-based segmentation of the heart in CT images," *Medical Imaging, IEEE Transactions on*, vol. 27, no. 9, pp. 1189–1201, 2008.
- [20] A. B. Redwood, J. J. Camp, and R. A. Robb, "Semiautomatic segmentation of the heart from CT images based on intensity and morphological features," in *Proceedings of SPIE*, 2005, vol. 5747, p. 1713.
- [21] R. Sammouda, J. Hassan, and M. Sammouda, "CT image analysis for early detection of lung cancer," *International Journal of Innovative Computing, Information, and Control*, vol. 4, no. 11, pp. 2847–2860, 2008.
- [22] R. SAMMOUDA, N. NIKI, and H. NISHITANI, "Segmentation of brain MR images based on neural networks," *IEICE TRANSACTIONS on Information and Systems*, vol. 79, no. 4, pp. 349–356, 1996.
- [23] S. Amatur, D. Piraino, and Y. Takefuji, "Optimization neural networks for the segmentation of magnetic resonance images," *Medical Imaging, IEEE Transactions on*, vol. 11, no. 2, pp. 215–220, 1992.
- [24] A. El-Baz, A. A. Farag, R. Falk, and R. La Rocca, "Detection, visualization and identification of lung abnormalities in chest spiral CT scan: Phase-I," in *International conference on biomedical engineering*, Cairo, Egypt, 2002, pp. 12–01.
- [25] D. T. Lin and C. R. Yan, "Lung nodules identification rules extraction with neural fuzzy network," in *Neural Information Processing, 2002. ICONIP'02. Proceedings of the 9th International Conference on*, 2002, vol. 4, pp. 2049–2053.
- [26] K. Kanazawa et al., "Computer-aided diagnosis for pulmonary nodules based on helical CT images," *Computerized medical imaging and graphics*, vol. 22, no. 2, pp. 157–167, 1998.
- [27] M. Sammouda, "Modification of the mean-square error principle to double the convergence speed of a special case of Hopfield neural network used to segment pathological liver color images," *BMC Medical Informatics and Decision Making*, vol. 4, p. 22, 2004.